



Engr. Abdul Wahab

Mechanical Design Engineer & Product Designer

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RESEARCH INTERESTS

Product Development Embedded Systems Finite Element Analysis (FEA)
Computational Fluid Dynamics (CFD) Autonomous Systems Mechanical Systems
DFM & DFA Simulations Mechanical Engineering Computer-Aided Design (CAD)
Computer-Aided Engineering (CAE) 3D Printing

PROFILE SUMMARY

As an innovative Mechanical Design Engineer with a comprehensive background in full-cycle Product Development and Mechanical Engineering, I am highly driven to get advanced knowledge through Master's program. My expertise encompasses the orchestration of sophisticated Mechanical and Autonomous Systems, leaning heavily on Computer-Aided Engineering (CAE), CAD, embedded systems, simulation and 3D Printing. By employing rigorous Simulations including Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) and integrating Embedded Systems, I consistently translate conceptual blueprints into robust, production-ready architectures. Having rigorously enforced DFM and DFA paradigms to streamline manufacturability, my overarching ambition is to fuse high-fidelity simulation insights with strategic engineering execution to pioneer highly optimized, future-ready technical systems.

WORK EXPERIENCE

June 2026 — September 2026

Mechanical Design Engineer

Dysol (Pvt) Ltd, Lahore, Pakistan

R&D Department — Key Responsibilities & Contributions

- Serving as a core Mechanical Design Engineer in the Research & Development department, focusing on innovative product development.
- Leading the end-to-end design and optimization of complex mechanical systems, including medical devices and fitness equipment.
- Conducting iterative engineering analysis, simulation, and validation to ensure functional performance and reliability.
- Applying DFM and DFA principles to bridge the gap between conceptual design and full-scale manufacturing.
- Collaborating across disciplines to integrate mechanical, fluidic, and electronic subsystems into cohesive, production-ready architectures.

August 2025 — Present

Mechanical Design Engineer

Upwork (National & International Clients)

- Delivered professional mechanical and product design services to national and international clients, applying rigorous engineering standards.
- Engineered precise 3D CAD models, complex assemblies, 2D technical drawings, and photorealistic renders for prototyping and manufacturing.
- Optimized components for consumer products and industrial applications, focusing on optimal material selection and manufacturing processes.

- Maintained a high degree of technical communication to translate client requirements into verifiable engineering parameters.

EDUCATION & TRAINING

September 2019 — October 2023

B.Sc. Mechanical Engineering

University of Engineering and Technology (UET), Peshawar — KPK, Pakistan

FINAL YEAR THESIS PROJECT

Design and Fabrication of Mechanism for Automated Installation of 18-Liter Water Bottle for Water Dispensers

Supervised by **Prof. Usman**

Thermodynamics Fluid Mechanics Machine Design CAD / CAM Manufacturing Processes
Heat Transfer Mechanics of Materials Engineering Dynamics

TECHNICAL SKILLS

CAD & Design

- SOLIDWORKS (3D / 2D Modeling)
- Parametric & Advanced Surface Modeling
- Technical Drawing with GD&T
- Photorealistic Rendering

Simulation Software

- ANSYS (Structural & Thermal)
- Altair EDEM (DEM Simulation)
- SolidWorks Simulation & Flow
- Motion Analysis

Engineering Analysis

- Finite Element Analysis (FEA)
- Computational Fluid Dynamics (CFD)
- Design for Assembly (DFA)
- Design for Manufacturability (DFM)
- Minitab (Data Analysis)

Manufacturing & Prototyping

- Additive Manufacturing (3D Printing)
- CNC Machining Optimization
- Injection Molding Prep
- Tolerance Stack-up Analysis

COURSES & CERTIFICATIONS

- **IELTS Academic** — Band 6.0 (Listening, Reading, Writing, Speaking)
- **Certified SOLIDWORKS Associate (CSWA)** — Dassault Systèmes
- **Pakistan Engineering Council (PEC)** — Registered Engineer
- **Lean Six Sigma Yellow Belt (LSSYB)** — udemy.com
- **Lean Six Sigma White Belt (LSSWB)** — udemy.com
- **Machine Learning in Python** — codanics.com
- **Introduction to Minitab** — simplilearn.com
- **CPR, AED & FIRST AID** — alison.com
- **ABC of Statistics** — codanics.com
- **AI ka Chilla 2026** — codanics.com
- **Python** — codanics.com

FEATURED PROJECTS

1. Fitness Product Development — Mechanical Dispensing System

Industry: Fitness & Consumer Products

Key Contributions:

- Designed mechanical assemblies involving gears, rack mechanisms, springs, sliding mechanisms, and functional interfaces.
- Developed solutions for multi-function operation within limited mechanical movement constraints.
- Performed multiple design iterations to improve mechanism reliability and manufacturability.
- Applied DFM and DFA principles for production-oriented design.
- Conducted motion analysis and mechanical validation.

Challenges Managed:

Optimizing complex mechanical interactions while maintaining compact design requirements. Achieved functional mechanism performance through iterative CAD refinement and simulation.

Tools Used:

SolidWorks (3D Modeling, Assembly Design, Motion Analysis), Altair EDEM (Simulation)

2. Medical Device Development — Controlled Flow Mechanical System

Industry: Medical Technology

Key Contributions:

- Designed mechanical components for controlled flow and operational safety.
- Developed assemblies involving valves, filters, actuators, and flow-control components.
- Applied mechanical design principles related to pressure management, thermal considerations, and fluid behavior.
- Implemented DFM and DFA methodologies for manufacturing readiness.
- Performed simulation-based design validation.

Challenges Managed:

Maintaining reliable mechanical operation under strict performance requirements. Improved design reliability through engineering analysis and simulation.

Tools Used:

SolidWorks (CAD Modeling & Motion Analysis), ANSYS (Simulation & Validation)

3. Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

Industry: Academic Research (Final Year Project)

Key Contributions:

- Designed the complete 3D CAD model for the automated mechanism.
- Developed a comprehensive solution to automate the manual lifting process, significantly reducing human effort.
- Engineered a customized lifting-to-adjustment sequence for seamless water bottle installation and atomization.
- Optimized the design by utilizing a minimum number of mechanisms to achieve cost-effective and reliable results.

Challenges Managed:

Designing a reliable lifting mechanism capable of handling 18-liter loads safely and efficiently. Balancing cost-effectiveness with robust mechanical performance through design optimization.

Tools Used:

SolidWorks (3D Modeling, Simulation, Animation, Motion Analysis)

AI FEATURED PROJECTS

Doc Agent — AI Research Assistant

[View Live App ↗](#)

An intelligent document analysis system built to make Research & Development incredibly easy. Users can upload any type of file and interactively ask questions to extract complex knowledge and information hidden within the documents instantly.

[Python](#) [Streamlit](#) [Large Language Models \(LLMs\)](#) [Document Parsing](#)

Visual Keyword Cloud Generator

[View Live App ↗](#)

An automated analytics application that enables professionals to generate dynamic visual keyword clouds from text, uploaded documents/files, professional work, projects, skills, or featured achievements. Instead of manual inspection, the app automatically analyzes content and presents high-frequency keywords in an intuitive, easy-to-understand visual format.

[Python](#) [Streamlit](#) [Text Processing](#) [Word Cloud Generation](#) [File Handling](#)
[AI-assisted Development](#) [App Deployment](#)

LANGUAGES

Language	Level
English	Proficient (B2)
Urdu	Fluent (C2)
Seraiki	Native (C2)

ACADEMIC
REFERENCES

Prof. Dr. Naveed Ahmad

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Prof. Muhammad Alam Zaib Khan

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